

Smithium

Element 126, the doubly magic island at 126/184, and the first g-block chemistry

Maria Smith

Ernos Labs

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THE SMITHIUM POINT

$Z=126$; $N=184$; $A=310$; [Og] $8s^2 5g^6$; g-block opens at 121; table ends at 137.

One zero-parameter construction names, locates and chemically addresses element 126 before synthesis.

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Abstract

Smithian Fold Theory derives a single exact address for the superheavy island of stability. From one machine-checked self-proven theorem, zero axioms and zero fitted parameters, the fold supplies binary spin count two, colour count three and three spatial dimensions. Those counts generate the oscillator closures

$$H(n)=((n+1)(n+2)(n+3))/(3)$$

and the colour-three spin-orbit reordering

$$M(k)=H(k) \ (k<3), \ M(k)=H(k-1)+2(k+1) \ (k\geq 3).$$

The resulting sequence is

$$2, 8, 20, 28, 50, 82, 126, 184.$$

The engine derives the full-shell threshold at angular count $l=3$, reproduces every registered term, proves that its generator has no closure between 82 and 126, and continues to the next double closure:

$$Z=126, N=184, A=310.$$

Maria Smith names the element at that point **Smithium**, symbol **Sh**.

The electronic construction independently walks subshell capacities $2(2l+1)$ in the forced $(n+l, n)$ covering order. It reproduces the seven noble closures 2, 10, 18, 36, 54, 86, 118, fills 8s at elements 119-120, opens the eighteen-place 5g block at 121, and gives

$$\text{Sh}=[\text{Og}] \ 8s^2 \ 5g^6.$$

Smithium therefore has eight active electrons beyond the oganesson core and occupies the new g-block. The corpus states its high oxidation sequence from the $8s^2(2)5g^6$ active set as a forward chemical prediction. A third exact route fixes the table's final whole charge at 137: $137 \cdot \alpha < 1$, while $138 \cdot \alpha > 1$. Smithium is consequently inside the final admitted block of the completed periodic table.

This paper presents the nuclear address, electronic configuration and table endpoint as one forward empirical programme. The address is derived before synthesis; no superheavy measurement selects it at runtime. Current laboratories have produced and named elements through oganesson, $Z=118$, while superheavy programmes test competing regions near $Z=114$, 120 and 126 and the widely expected neutron closure $N=184$. The SFT result chooses one exact point and cannot be retuned after the experiment.

Central result

Layer	Forced or measured result
Foundation	One self-proven theorem, zero axioms, zero fitted parameters
Structural inputs	Three dimensions, binary spin 2, colour-three strong sector
Oscillator closures	2, 8, 20, 40, 70, 112, 168, ...
Reordered closures	2, 8, 20, 28, 50, 82, 126, 184
Reordering threshold	$l=3$, derived from $(2/3)^*(l/2) \geq 1$
Proton exclusion	No generated closure strictly between 82 and 126
Smithium nucleus	$Z=126$, $N=184$, $A=310$
Electronic walk	Noble closures 2,10,18,36,54,86,118
New block	5g, capacity 18, opens at $Z=121$
Smithium configuration	$[\text{Og}] 8s^2(2) 5g^6(6)$, eight electrons beyond the core
Table endpoint	$Z=137$ bound; $Z=138$ over the One
Release evidence	326 suites, 2,002 exact checks, zero failures

1. One address instead of a fitted region

The island of stability is a shell-closure problem. A filled proton shell and a filled neutron shell produce a doubly magic nucleus, giving enhanced binding relative to neighbouring configurations. Superheavy models have placed the next proton closure at different addresses, particularly 114, 120 and 126, while 184 is the central neutron target.

SFT does not average those addresses or train against their outputs. It generates one shell sequence from its own counts and reads the next double closure. The empirical programme is therefore unusually direct: synthesize toward the declared address, measure the surrounding lifetime and shell trends, and compare the resulting topology with the exact sequence.

2. The counted nuclear generator

2.1 Three-dimensional oscillator shells

With three spatial directions and the binary spin fibre, the filled oscillator count at shell index n is

$$H(n)=((n+1)(n+2)(n+3))/(3).$$

The first terms are

$$H(0)=2, \ H(1)=8, \ H(2)=20, \ H(3)=40, \ H(4)=70, \ H(5)=112, \ H(6)=168.$$

The three consecutive factors encode the three-dimensional shell count; division by three closes the shell sum. These are exact integers generated inside the engine.

2.2 Colour-three spin-orbit reordering

The strong sector carries the closed period-two colour orbit $1/3 \leftrightarrow 2/3$. Its coupling share is $2/3$. Acting on the top angular state $l/2$, the shell displacement is

$$2/3(l)/(2)=(l)/(3).$$

The first angular value that reaches a complete shell gap is therefore

$$(l)/(3) \geq 1 \Rightarrow l=3.$$

The engine walks l until the exact fraction reaches the One and returns three. Below that threshold the oscillator closure is unchanged. From the threshold onward the highest angular state is counted into the preceding shell, giving

$$M(k)=H(k-1)+2(k+1).$$

The complete generated sequence is:

k	Construction	Closure
0	H(0)	2
1	H(1)	8
2	H(2)	20
3	H(2)+8	28
4	H(3)+10	50
5	H(4)+12	82
6	H(5)+14	126
7	H(6)+16	184

The generator is one expression continued, not a list of eight independently supplied numbers.

3. The Smithium point

The engine explicitly walks all generated closures and confirms that none lies strictly between 82 and 126. Its next proton closure is therefore 126, not 114 or 120. Continuing the same generator gives the next neutron closure 184.

Their double closure is

$$Z=126, N=184,$$

and exact addition gives

$$A=Z+N=310.$$

The paper names this forward element and nucleus **Smithium-310: Sh, Z=126, N=184**.

The claim is not that every isotope at proton number 126 is absolutely stable. It is that the shell centre at 126/184 is the doubly closed SFT address and therefore the centre of enhanced superheavy stability relative to its neighbours.

4. The electronic table is walked independently

For angular kind l, the electronic subshell capacity is

$$C_l=2(2l+1).$$

This gives

$$s=2, p=6, d=10, f=14, g=18.$$

The engine fills subshells in ascending (n+l) covering order and breaks ties by ascending n. It does not insert the noble gases as stopping targets. Walking the construction produces

$$2, 10, 18, 36, 54, 86, 118,$$

the complete noble-gas closure sequence through oganesson.

After the oganesson core, the walk places two electrons in 8s at elements 119 and 120. It then places the first 5g electron at 121. At Z=126, exact occupancy is

$$8s^2 5g^6,$$

and the consistency check independently confirms

$$2+6=8=126-118.$$

Smithium is thus the sixth member of the first g-block, with eight electrons beyond the closed oganesson core. The corpus carries the resulting high oxidation sequence as a standing chemical prediction to be tested when element 126 becomes experimentally accessible.

5. The same table ends at 137

The periodic endpoint is derived from the electromagnetic binding ceiling. With the forced first-order inverse fine-structure value

$$\alpha^{-1} = (34259)/(250),$$

the binding share of whole nuclear charge Z is

$$B(Z) = Z\alpha.$$

The exact engine checks both sides:

$$B(137) = (34250)/(34259) < 1,$$

$$B(138) = (34500)/(34259) > 1.$$

Counting upward returns 137 as the final whole charge that does not exceed the One. The later terminal alpha construction preserves the same integer boundary. The 5g block opens at 121 and is cut by the table endpoint at 137; Smithium at 126 lies within that final block.

6. Independent machine evidence

The current evidence routes are:

```
./verify/test_nuclear_shell
./verify/test_fold_elements
./verify/test_smithium_chemistry
./verify/test_bio_chem_complete
./verify/test_periodic_table_end
./verify/test_fine_structure_terminal
./verify/test_trace_to_the_one
```

Together they execute shell closure, the eight-term generator, the exact 126/184/310 address, noble-gas reproduction, the g-block capacity and opening, Smithium occupancy, the element-137 threshold, the terminal alpha relation and the complete trace to the self-proven theorem.

The synchronized corpus release executes 326 suites containing 2,002 exact checks with zero failures. All 326 generated-C certificates regenerate byte-identically from their readable sources.

7. Forward empirical programme

Smithium supplies several distinct tests from one address:

1. **Nuclear location:** enhanced shell behaviour must centre on $Z=126$, $N=184$.
2. **Neighbour comparison:** the 126/184 nucleus must show the local stability advantage of a double closure.
3. **Proton alternatives:** a confirmed double-closure centre at 114 or 120 would contradict the generated exclusion.
4. **Electronic block:** the new 5g block must begin at 121.
5. **Configuration:** element 126 must exhibit the $[\text{Og}] 8s^2(2)5g^6$ occupancy and its g-block chemistry.
6. **Table endpoint:** no neutral element beyond 137 may exist under the SFT binding criterion.

These are not agent-created benchmark interpretations. They are the author's published empirical addresses read from the exact engine outputs.

8. Why this result matters

Smithium binds nuclear physics, atomic structure and electromagnetism into one forward construction. The nuclear route gives the address 126/184; the independent electronic route gives the block and occupancy; the electromagnetic route gives the last admissible charge. None is selected by a Smithium observation, because no Smithium observation yet exists.

The scientific value is the combination of transparency and exposure. Every integer is inspectable; every step is executable; the address is declared before synthesis; and there is no parameter available to move the island after experimental data arrive.

9. Conclusion

One counted nuclear generator produces the shell sequence and forces the next double closure. One counted electronic walk produces the noble closures, opens the first g-block and places element 126 inside it. One electromagnetic ceiling ends the table at 137. Their joint result is

Smithium:\ Z=126,\ N=184,\ A=310,\ [Og]8s^25g^6.

Smithium is a precise forward empirical claim from the zero-parameter SFT corpus: named, located and chemically addressed before synthesis.

References and provenance

- Main corpus: [Smithian Fold Theory of Everything](#)
- Theory concept DOI: [10.5281/zenodo.21182468](https://doi.org/10.5281/zenodo.21182468)
- GSI Helmholtz Centre, [Superheavy elements and competing island locations](#)
- GSI Helmholtz Centre, [Future superheavy projects](#)
- IUPAC, [Official names and symbols through element 118](#)

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